1

36/36

300 students take part in a competition.
The table shows information about their scores.

	mid v	15	35	50	65	85
Score (s)	$0 < s \leqslant 30$	$30 < s \leqslant 40$	$40 < s \leqslant 60$	$60 < s \leqslant 70$	$70 < s \leqslant 100$
Frequency	24	70	88	76	42
C.W	30	10	20	10	30
δd :	0.8	7	4.4	7.6	1.4

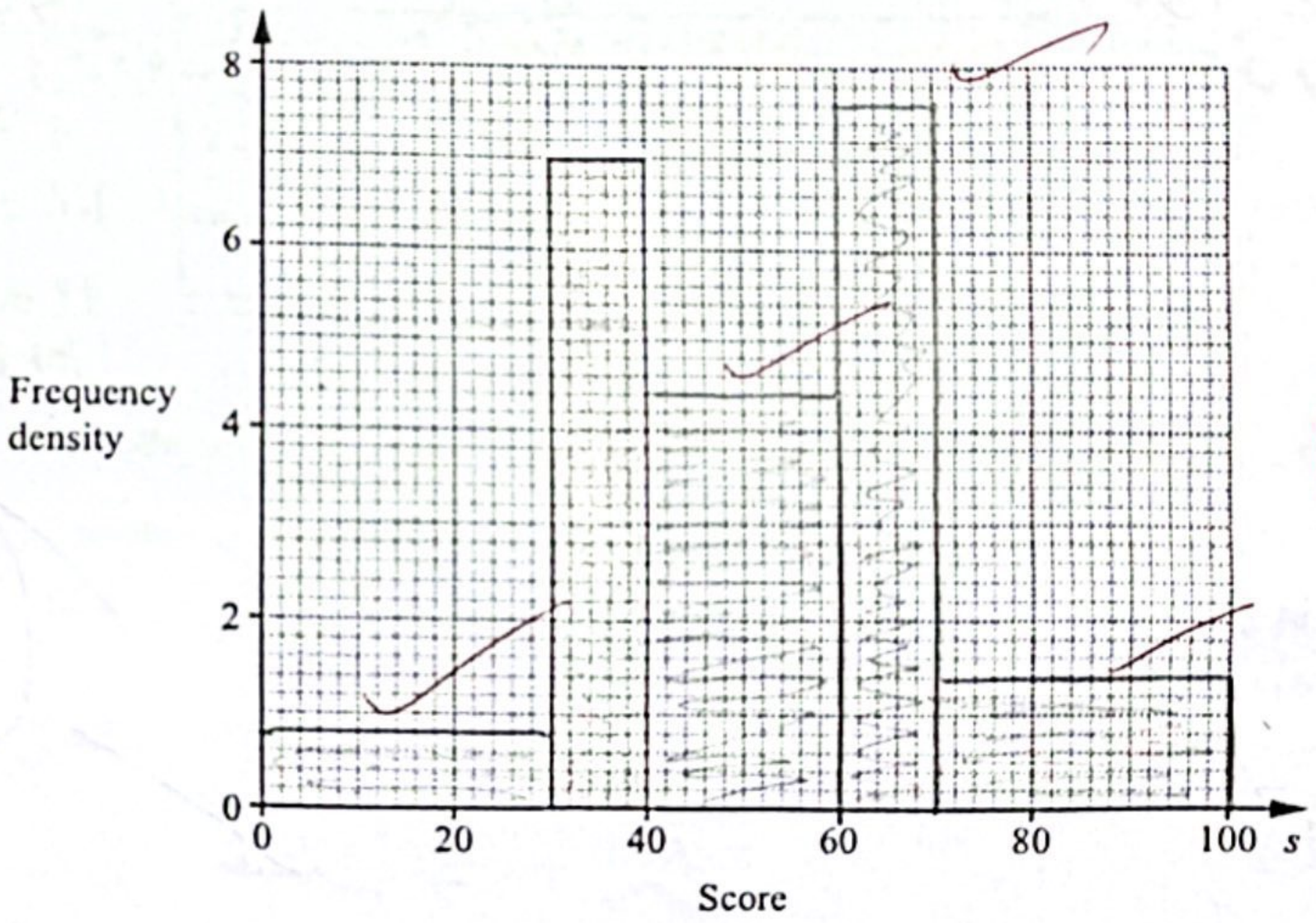
(a) Calculate an estimate of the mean score.

$$\frac{15\times24 + 35\times70 + 50\times88 + 65\times76 + 85\times42}{24+70+88+76+42} = \frac{15\,720}{300} = 52.4$$

52.4 [4]

(b)

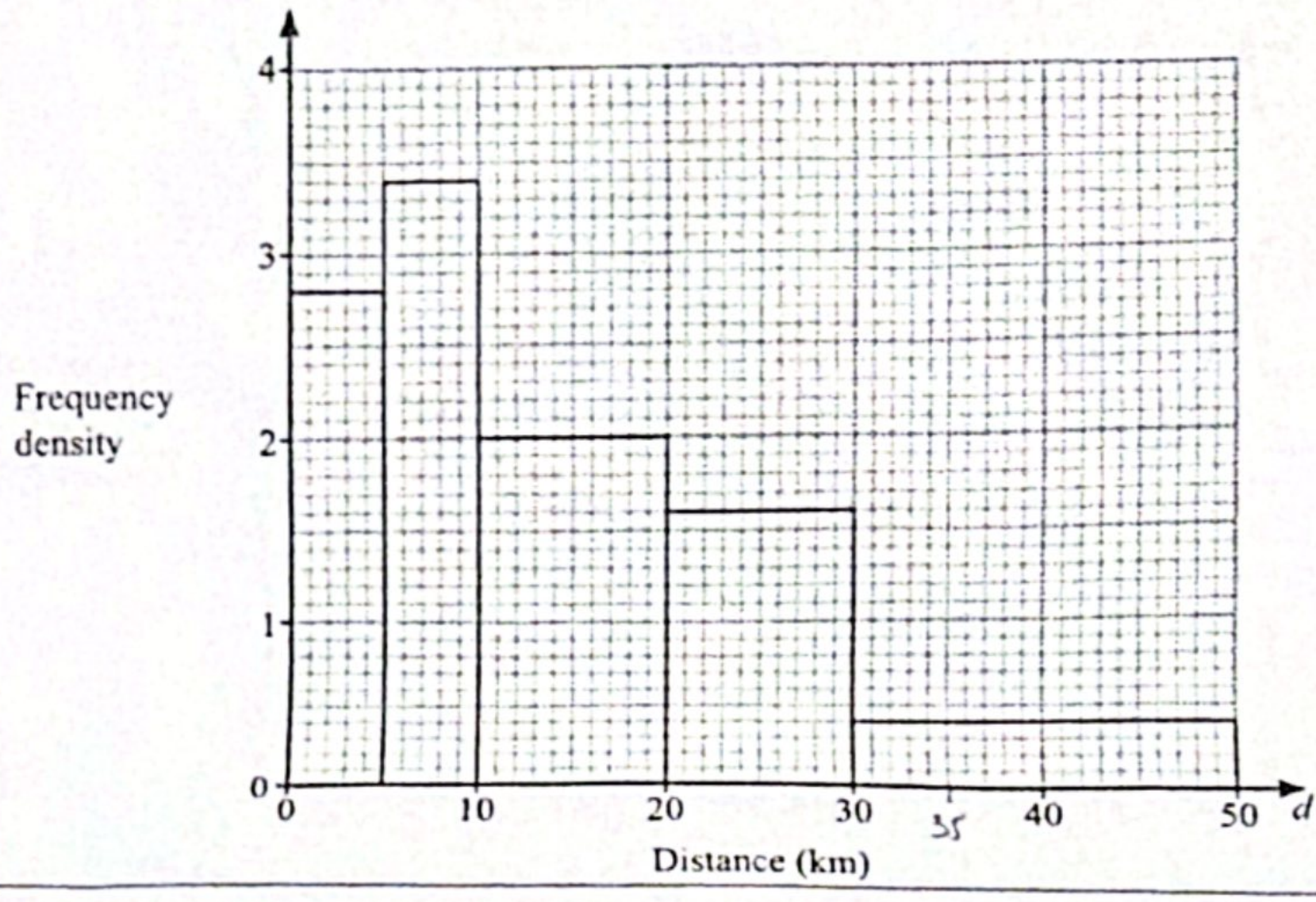
$\delta d = \frac{\delta}{C.W}$



On the grid, complete the histogram to represent the data in the table. [3]

2

The histogram shows information about the distance some people travel to work.



(a) Use the histogram to complete the table.

Distance (d km)	Number of people
$0 < d \leqslant 5$	14
$5 < d \leqslant 20$	37
$20 < d \leqslant 50$	24

[3]

(b) The people who travel a distance greater than 35 km to work have a car allowance.

Calculate an estimate of the number of these people who have a car allowance.

4

..... [1]

3

The table shows some information about the marks scored by a group of students in a test.

Test mark	4	5	8
Frequency	2	4	n

The mean mark is 6.

Work out the value of n.

$$\frac{4\times2 + 5\times4 + 8\times n}{2+4+n} = 6$$

$$8 + 20 + 8n = 6(6+n)$$

$$28 + 8n = 36 + 6n$$

$$8n - 6n = 36 - 28$$

$$2n = 8$$

$$n = 4$$

$$6 = \frac{8+20+8n}{6+n}$$

$$36 + 6n = 28 + 8n$$

$$8 = 2n$$

$$4 = n$$

3

$n =$ ..... 4 ..... [3]

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4

There are 15 giraffes in a group.
The table gives information about the heights of the 15 giraffes.

One giraffe has a height of 2.6 m
No giraffe is shorter than 2.5 m
The range of heights for the 15 giraffes is 2.3 m
More than 3 giraffes have the same height
The modal height for the giraffes is 3.9 m

The stem-and-leaf diagram shows information about the height of 9 of these giraffes.

2	5 6
3	2 7 7 9 9 9 9
4	1 1 4 5 7 8

Key: 4|1 represents a giraffe height of 4.1 m

Use the information in the table to complete the stem-and-leaf diagram for the group of 15 giraffes.

4 [3]

5

The median of six numbers is 61.
Five of the numbers are 24, 43, 58, 71 and 85.

Work out the sixth number.

$\frac{6+1}{2} = 3.5$

64 [1]

24, 43, 58, 64, 71, 85

24, 43, 58, x, 71, 85

$61 = \frac{58+x}{2}$

$122 = 58 + x$

$64 = x$

$\frac{58+x}{2} = 61$

$58 + x = 122$

$x = 64$

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6 The following are the numbers of goals scored by 11 players in a football tournament.

2, 3, 4, 4, 5, 5, 5, 6, 7, 7, 8

(a) Calculate the mean number of goals scored. [1]

$\frac{2+3+4+4+5\times3+6+7\times2+8}{11}$

$= 5.0909$

$\times 5.09$

(b) Write down the mode. [1]

$= 5$

(c) Write down the median. [1]

$\frac{11+1}{2} \rightarrow 6$

median $= 5$

(d) Calculate the range. [1]

$8-2=6$

6

(e) Calculate the interquartile range. [2]

$Q3 = \frac{75}{100} \times 11 = \frac{825}{100} = 8.25$

$Q1 = \frac{25}{100} \times 11 = \frac{275}{100} = 2.75$

$Q3 - Q1 = IQR$

$8.25 - 2.75 = IQR$

$IQR = 5.5$

2, 3, 4, 4, 5, 5, 5, 6, 7, 7, 8

1 2 3 4 5 1 2 3 4 5

4 7

$7-4=3$

$IQR = 3$

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7 The table shows the number of books read by a group of students during one month.

Use the table to answer the questions below.

Number of books, x	0	1	2	3	4
Frequency, f	3	7	9	5	1

(a) Calculate the mean number of books read. [2]

$\frac{0\times3+1\times7+2\times9+3\times5+4\times1}{3+7+9+5+1} = \frac{44}{25} = 1.76$

(b) Write down the mode. [1]

= 2

4

(c) Write down the median. [1]

3+7=10+9=19 → 13 ν αω ο

= 2

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8. Matilda records the distances that 80 different cars can travel with a full tank of fuel.
The table shows this information.

mid v 275 350 410 435 475

Distance (d km)	$250 < d \leqslant 300$	$300 < d \leqslant 400$	$400 < d \leqslant 420$	$420 < d \leqslant 450$	$450 < d \leqslant 500$
Frequency	7	13	19	21	20

cw : 50 100 20 30 50

(i) Write down the class interval that contains the median.

$420 < d \leq 450$

420 ..... $< d \leqslant$ 450 ..... [1]

(ii) Calculate an estimate of the mean.

$$\frac{275\times7 + 350\times13 + 410\times19 + 435\times21 + 475\times20}{7+13+19+21+20} = \frac{32900}{80} = 411.25$$

411.25 ≈ 411

411.25 ≈ 411 ..... km [4]

(iii) A histogram is drawn to show the information in the table.
The height of the bar for the interval $250 < d \leqslant 300$ is 2.8 cm.

Calculate the height of the bar for each of the following intervals.

8

ans: 0.14, 0.13, 0.95, 0.7, 0.4

0.14 = 2.8 0.14 = 2.8
0.13 = x 0.95 = x
0.364 = 0.14x 2.66 = 0.14x
2.6 = x 19 = x

0.14 = 2.8
0.7 = x
1.96 = 0.14x
19 = x

$300 < d \leqslant 400$ 2.6 ..... cm

$400 < d \leqslant 420$ 19 ..... cm

$420 < d \leqslant 450$ 14 ..... cm [3]

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